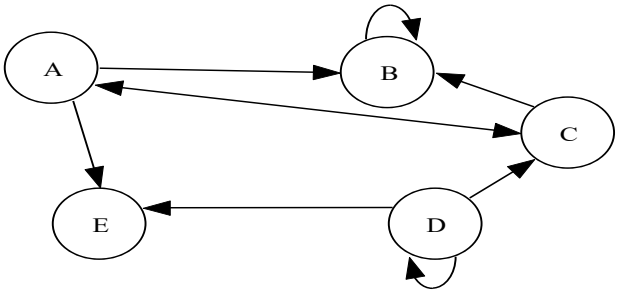
Compiled Problems – Intermediate Division

1. **00-01 C3 Graph Theory**

   How many paths of length 2 exist in the following graph?



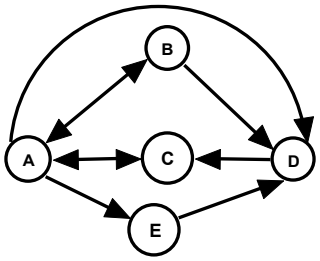
2. **00-01 C3 Graph Theory**

   Which of the following paths are cycles?

   A) AB
   B) ABCA
   C) ABCABA
   D) AAA
   E) ABCDE

3. **01-02 C3 Graph Theory**

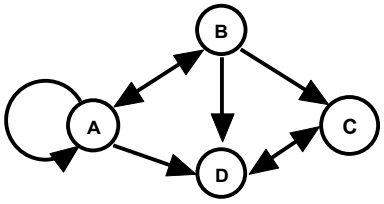
   Which of the following are simple paths in the following diagram?



   A) ECAD
   B) BCABD
   C) EDCAB
   D) CADCB
   E) None of the above

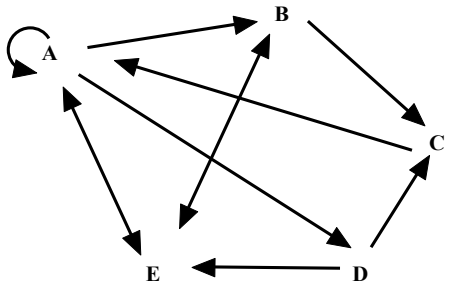
4. **01-02 C3 Graph Theory**

   Write the adjacency matrix for the following graph.



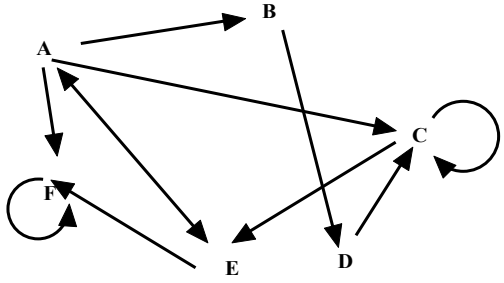
5. **02-03 C3 Graph Theory**

How many different cycles in the following directed graph contain vertex A?



6. **02-03 C3 Graph Theory**

How many paths of length 2 exist in the following directed graph?



7. **03-04 C3 Graph Theory**

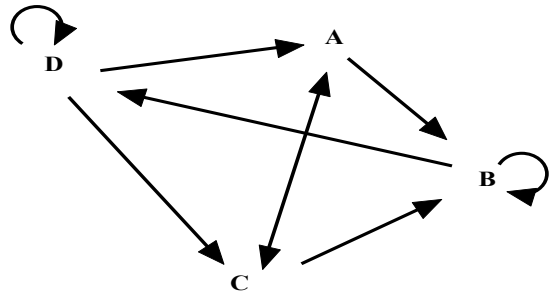
Draw the directed graph with vertices {A, B, C, D, E} and edges {AB, DA, CE, DE,AE, BC, CA}.

8. **03-04 C3 Graph Theory**

Given the directed graph with vertices {A, B, C, D, E} and edges {AB, DA, CE, DE, AE, BC, CA}, how many different cycles exist in the directed graph?

9. **04-05 C3 Graph Theory**

How many cycles exist in the following directed graph?

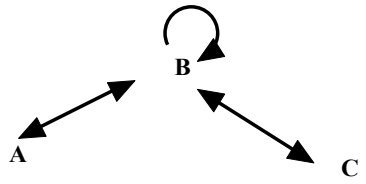


10. **04-05 C3 Graph Theory**

Given the vertices {A, B, C, D, E} and the edges {AB, AC, CB, DE, EC, BE}, draw the directed graph.

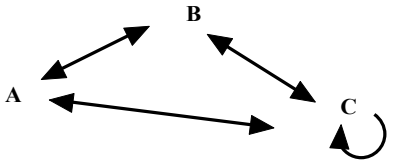
11. **05-06 C3 Graph Theory**

Given the following directed graph find the total number of different paths of length 2:



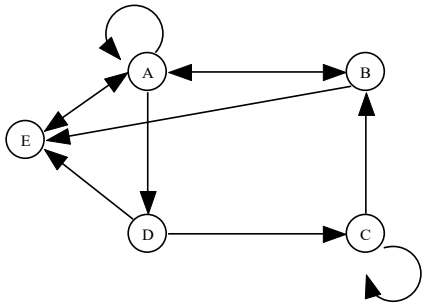
12. **05-06 C3 Graph Theory**

How many different cycles are in the directed graph below?



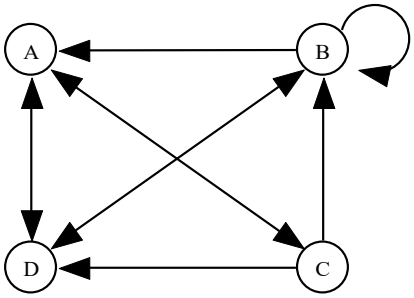
13. **06-07 C3 Graph Theory**

How many different cycles starting at vertex A exist in the directed graph?



14. **06-07 C3 Graph Theory**

How many paths of length 3 exist from vertex A to vertex D in this directed graph?



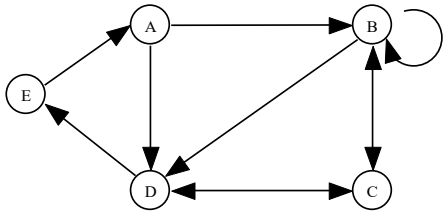
15. **07-08 C3 Graph Theory**

Given the adjacency matrix, draw the directed graph.

	A	B	C	D
A	1	0	1	1
B	0	1	1	0
C	1	0	1	0
D	1	1	1	1

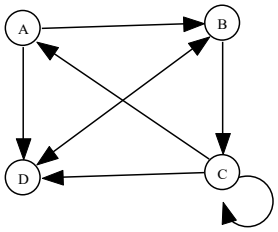
16. **07-08 C3 Graph Theory**

How many paths of length 2 exist from vertex B in this directed graph?



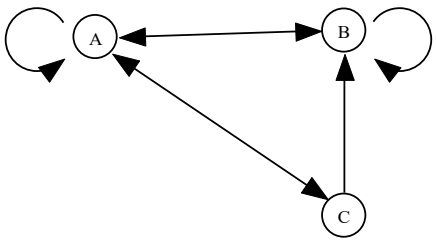
17. **08-09 C3 Graph Theory**

How many paths from B of length 3 exist in the following directed graph?



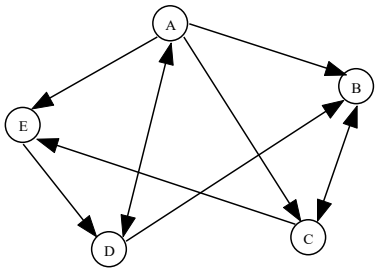
18. **08-09 C3 Graph Theory**

How many different cycles are in the following directed graph?



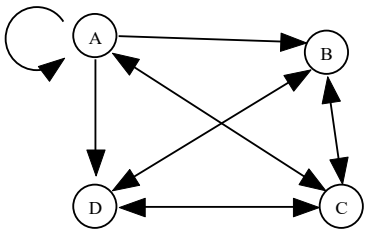
19. **09-10 C3 Graph Theory**

How many pairs of vertices do not have paths of length three?



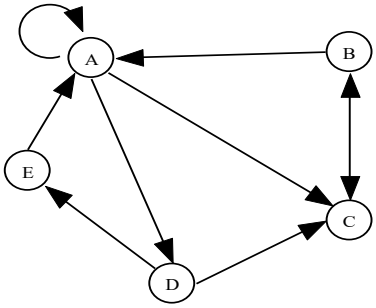
20. **09-10 C3 Graph Theory**

How many cycles are in the following directed graph from vertex C?



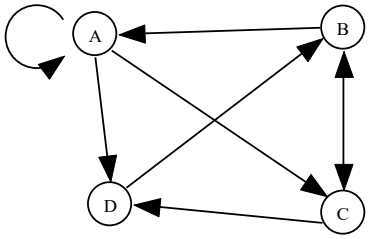
21. **10-11 C3 Graph Theory**

Draw the adjacency matrix for the following directed graph.



22. **10-11 C3 Graph Theory**

How many paths of length 3 exist in the following directed graph?

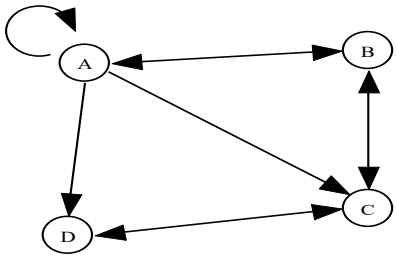


23. **11-12 C3 Graph Theory**

Draw the directed graph with the following vertices {A, B, C, D, E} and directed edges {AB, BB, CD, EA, DE, AD, CE, CB, CA, DB}.

24. **11-12 C3 Graph Theory**

How many paths of length 2 exist in the directed graph at the right?

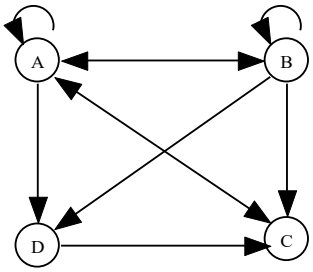


25. **12-13 C3 Graph Theory**

Given the directed graph with vertices A, B, C, D, and E and directed edges AB, AC, AD, BA, CD, EA, EC, CE, BD, and BE. How many pairs of vertices have no directed edge?

26. **12-13 C3 Graph Theory**

How many paths of length 3 exist in the following directed graph?

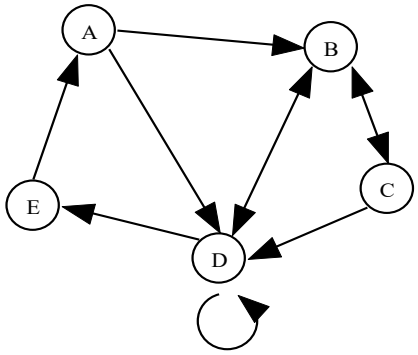


27. **13-14 C3 Graph Theory**

Given the directed graph with vertices A, B, C, and D, and directed edges AB, BD, CD, AC, CB, CA, DA, and AD. Draw the adjacency matrix represented by the directed graph.

28. **13-14 C3 Graph Theory**

How many paths of length 2 exist in the following directed graph?



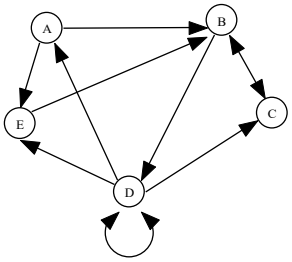
29. **14-15 C3 Graph Theory**

Given the following adjacency matrix, draw the directed graph.

$$\begin{vmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{vmatrix}$$

30. **14-15 C3 Graph Theory**

Which starting vertex has the most paths of length 2?



31. **15-16 C3 Graph Theory**

Given the following adjacency matrix, how many lines between vertices are bidirectional, i.e. arrowhead at both ends?

$$\begin{vmatrix} 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{vmatrix}$$

32. **15-16 C3 Graph Theory**

Which directed pair(s) of vertices do not have paths of length 3?

